

Comprehensive Academic Review and Analysis: Decentralized Swarm Autonomy Framework for Cyber-Physical Systems

1. Executive Summary and Overall Quality Assessment

The manuscript titled "Decentralized Swarm Autonomy Framework for Cyber-Physical Systems under Communication and Energy Constraints" presents a highly ambitious, structurally complex, and well-engineered software architecture designed to simulate decentralized multi-agent robotic swarms. The core premise of the work revolves around subjecting autonomous agents to severe epistemic uncertainty—often characterized within the manuscript as the "Fog-of-War"—which encompasses stochastic packet dropping, asynchronous communication delays, and irreversible thermodynamic energy depletion. By architecturally restricting agents to strictly localized, delayed belief buffers, the framework aims to evaluate the resilience of various swarm coordination algorithms, including Sequential Single-Item (SSI) auctions, localized Voronoi coverage, and gossip-based consensus protocols. To maintain system stability during adaptive heuristic mutations, the architecture introduces a mechanism termed a Θ_{safe} box-clamping projector coupled with exponential moving average (EMA) smoothing.

Evaluated strictly against professional, peer-reviewed research standards in the fields of control theory, multi-agent systems, and cyber-physical systems (CPS), the overall quality of the manuscript represents a sharp dichotomy. On one hand, the software engineering, system architecture, and simulation design are exceptional, demonstrating a profound understanding of deterministic execution, event-driven programming, and the practical challenges of distributed systems. On the other hand, the control-theoretic foundations and the mathematical formalisms supporting the framework exhibit severe theoretical deficiencies. The ambition of the conceptual framework significantly outpaces the maturity of the mathematical formulations. The manuscript leverages highly advanced domain terminology—such as spectral connectivity, epistemic boundaries, thermodynamic energy cascades, and safe-set projection—but repeatedly falls back on heuristic, overly simplistic mathematical models that fail to substantiate the grand claims made in the abstract and introduction.

Ultimately, the manuscript currently reads as a sophisticated and exceptionally well-written engineering simulation manual rather than a mature, theoretically rigorous contribution to decentralized control literature. The writing is persuasive, the formatting is professional, and the systemic motivation is flawless, yet the analytical proofs required by premier IEEE venues are absent.

2. Articulation of the Research Problem and

Motivation

The research problem is exceptionally well-motivated, representing one of the most compelling aspects of the manuscript. The authors accurately identify a critical vulnerability in contemporary multi-agent simulation literature: the pervasive, almost systemic reliance on idealized, continuous-time feedback loops and zero-latency, global communication graphs. The manuscript's assertion that algorithms developed under these perfect-information assumptions undergo catastrophic failure modes when exposed to physical hardware realities—such as radio frequency (RF) interference, topological fragmentation, and unpredictable packet loss—is empirically sound and highly relevant to modern cyber-physical system deployments. Furthermore, the explicit modeling of irreversible thermodynamic energy depletion adds a layer of practical realism rarely addressed in standard consensus literature. In active physical deployments, energy is indeed a monotonically decreasing resource. The manuscript correctly postulates that when an agent fully depletes its energy and drops out of the network, surviving agents must autonomously compensate by expanding their spatial coverage radius and boosting transmission attempts to maintain connectivity. The identification of this phenomenon as an "inverse energy cascade," where localized physical adaptation burns exponentially more energy across the remaining swarm and triggers rapid network fragmentation, demonstrates a deep, intuitive grasp of the compounding failures inherent in resource-constrained cyber-physical systems. The motivation is clear, grounded in physical reality, and eloquently argued.

3. Genuine Novelty and its Articulation

The novelty of the framework is clearly articulated, particularly in Section I and the accompanying Table I, but the genuineness of this novelty is mixed, requiring a nuanced evaluation.

The primary and most genuine novelty lies in the architectural integration of these constraints into a deterministic, bit-exact reproducible Discrete-Event Simulation (DES). Forcing a strict epistemic boundary where agents operate exclusively on stale neighbor data without any back-channel access to global state arrays is a highly commendable engineering design choice. Many existing multi-agent simulators inadvertently leak global state information to agents to simplify collision avoidance or path planning. By explicitly decoupling the physical simulation physics from the decentralized agent logic, the authors have created a highly rigorous testbed for evaluating algorithms under true partial observability.

However, the algorithmic and mathematical components claimed as novel are largely derivative, heuristic, or overly simplistic, which undermines the manuscript's broader claims of theoretical innovation: First, the introduction of a "Passive Regime Detection" layer utilizing local telemetry proxies to drive an automaton is standard practice in hybrid control systems and behavioral robotics. While well-implemented here, it does not constitute a fundamental advancement in the state-of-the-art. Second, the "Stability-Constrained Adaptation" (Θ_{safe}) is described throughout the manuscript as a "safe-set projection." In modern nonlinear control theory, safe-set projection implies the use of Control Barrier Functions (CBFs), artificial potential fields, or invariant set theory to mathematically guarantee that a system's state remains within a predefined safe manifold. In this manuscript, the mechanism is simply implemented as a static maximum/minimum box-clamping function. Truncating a parameter to a minimum or maximum bound is a fundamental software engineering safeguard, not a novel, mathematically bounded

control-theoretic stability manifold. Third, the manuscript claims novelty in utilizing distributed spectral estimation for assessing network connectivity. While the concept of distributed estimation of algebraic connectivity is a vibrant research area, the manuscript entirely fails to provide the actual distributed estimation algorithm, instead presenting the centralized mathematical definition. Therefore, the novelty is genuine strictly within the domain of simulation software architecture for evaluation, but the paper severely overstates its novelty regarding fundamental distributed control algorithms.

4. Methodological and Technical Soundness

The systemic workflow, software architecture, and discrete-event pipeline are conceptually sound, very clearly articulated, and highly scalable. The separation of the system into an Orchestration Layer (managing the DES Kernel, Min-Heap Event Queue, and Communication Engine) and a Decentralized Agent Logic (housing the Telemetry Buffer, Hybrid Supervisor, and Heuristic Adapter) is the strongest technical aspect of the paper. This architecture explicitly prevents data races and strictly enforces the epistemic boundary. The data flow detailed in the sequence diagram maps a logical and computationally efficient pipeline.

The decentralized Sequential Single-Item (SSI) auction (Algorithm 2) is logically constructed for a delayed network. In a zero-latency environment, auctions can be cleared instantaneously. However, under "Fog-of-War" constraints, bid packets are frequently dropped or delayed. By forcing agents to bid based on localized distance and inverse energy margins, and crucially, incorporating a time-delayed AUC_RESOLVE event to handle asynchronous bid comparisons from neighbors, the workflow demonstrates a sophisticated understanding of distributed state resolution. If an agent does not receive a superior bid within the Δt_{auc} window, it claims the task, updating its Local Belief Buffer. This minimizes bandwidth consumption and preserves decentralized autonomy under contested network conditions.

Despite these architectural strengths, the underlying control methodology harbors deep technical flaws that compromise the theoretical soundness of the framework. The most egregious of these is the "Box-Clamping Fallacy." The manuscript claims that Algorithm 1 intercepts parameter mutations via simple max/min bounds, thereby acting as a "deterministic safety leash, ensuring that behavioral mutations... never force the swarm outside of a mathematically bounded stability manifold." This assertion represents a fundamental misunderstanding of dynamical systems stability. Stability is governed by the eigenvalues of the system matrix, the margins of the adaptation rates, and the energy of the system states, not merely the static bounds of a parameter. A control parameter can be strictly constrained within a "safe box" but oscillate rapidly and unpredictably from the minimum to the maximum bound at a high frequency. Such bang-bang behavior injects high-frequency noise into the drone flight controller, which can rapidly saturate actuators and destabilize the physical UAV. While the Exponential Moving Average (EMA) smoothing mechanism mitigates sudden impulse shocks, the claim that static clamping "mathematically bounds" the swarm without risking kinematic destabilization is theoretically unsupported and factually incorrect from a control engineering perspective.

Furthermore, the methodology fails to account for asymmetric epistemic boundaries in spatial partitioning. The framework models probabilistic packet dropping via a Bernoulli distribution. In such Random Geometric Graphs (RGG), the adjacency matrix becomes inherently directed and asymmetric; Agent i might successfully receive Agent j 's telemetry packet, but Agent j might drop Agent i 's packet due to localized interference. The methodology relies on localized Voronoi

tessellation for spatial coverage. However, under asymmetric communication, local Voronoi partitions will conflict. If Agent i knows about Agent j , it shrinks its Voronoi cell to share the space. If Agent j does not know about Agent i , it expands its cell, assuming the space is empty. This asymmetry leads to overlapping spatial territories, unassigned regions, and potential physical collisions, an issue well-documented in distributed coverage literature. The manuscript remains entirely silent on how the framework resolves these asymmetric overlaps, leaving a critical gap in the technical methodology.

5. Mathematical Rigor and Consistency Analysis

The mathematical rigor is unequivocally the weakest aspect of the manuscript. The formulations provided are heuristic, logically inconsistent with the paper's own architectural claims of decentralized autonomy, and entirely lack the requisite stability proofs, Lyapunov analyses, or frequency-domain bounds expected in professional IEEE literature. A rigorous deconstruction of the presented equations reveals multiple foundational errors.

5.1 Critique of the Stochastic RGG Adjacency Model

Equation 1 defines the stochastic communication edges, stating that the effective transmission radius is $\overline{R}_{tx} = R_{tx} - \omega_{env}$. This formulation implies that environmental attenuation (ω_{env}) is a static, linear geometric subtraction from the physical transmission radius. In actual radio frequency physics, signal-to-noise ratio (SNR) decays following an inverse-square law (or higher-order path loss models in contested environments), and interference is highly non-linear. While a linear subtraction might be acceptable as a highly simplified simulation proxy, coupling this with a uniform Bernoulli drop probability (p_{drop}) drawn from a continuous uniform distribution $\mathcal{U} \sim \text{Uniform}(0,1)$ oversimplifies the spatial correlation of jamming fields. In real electronic warfare scenarios, jamming creates spatially contiguous zones of high packet loss, rather than uniformly distributed random drops across the entire coordinate space.

5.2 The Distributed Spectral Connectivity Paradox

Equation 2 represents a fatal mathematical and logical flaw in the manuscript. The equation defines the distributed spectral connectivity (λ_2) using the standard centralized Rayleigh quotient definition of the Fiedler value, where $\lambda_2(L) = \min_{x \perp \mathbf{1}, x \neq 0} \frac{x^T L x}{x^T x}$. The Fiedler value is a global topological metric dependent on the eigenvalues of the entire global graph Laplacian matrix (L). The manuscript's central architectural thesis is that agents are strictly restricted from accessing global adjacency matrices due to the enforced epistemic boundary. However, the authors fail to provide any distributed estimation dynamics required to compute λ_2 locally.

The academic literature clearly establishes that distributed estimation of algebraic connectivity is a highly complex problem requiring sophisticated mechanisms, such as decentralized power iterations, PI consensus filters, or distributed observers to track the second smallest eigenvalue over time. Simply stating the global centralized definition and claiming that the hybrid supervisor computes a "scaled local proxy" leaves a massive mathematical void. How exactly is $\hat{\lambda}_2$ calculated by an isolated agent with only stale information from immediate neighbors? This omission invalidates the core regime-shifting mechanism demonstrated in the

experiments, as it implies the simulation kernel might secretly be calculating λ_2 globally and feeding it to the agents, which directly violates the paper's primary claim of strict decentralized fog-of-war constraints.

5.3 Linear Thermodynamic Energy Decay

Equation 3 models discrete thermodynamic energy decay through a monotonic, linear subtraction: $E_i(t_{k+1}) = E_i(t_k) - (\gamma_m d_{\text{mov}} + \gamma_e N_{\text{msg}} + \gamma_t \Delta t)$. While this linear formulation is sufficient for high-level finite state machines, thermodynamic decay in physical UAVs is inherently non-linear. For instance, battery discharge follows non-linear voltage curves characterized by Peukert's Law, and aerodynamic drag scales with the square of the velocity. More importantly, the introduction claims the framework models "thermodynamic energy cascades." In a true cascade, as nodes die, surviving nodes must expand their communication ranges to bridge topological gaps. RF transmission power scales exponentially (or at minimum quadratically) with distance. Therefore, as the swarm shrinks, the energy consumption per remaining agent should accelerate exponentially. A purely linear energy formulation cannot accurately model the complex, cascading failure dynamics the paper claims to solve, representing a disconnect between the theoretical narrative and the mathematical execution.

5.4 Flawed Discrete-Time Consensus Bounds

Equations 7 and 8 define the asynchronous state consensus update. To guarantee stability and prevent oscillatory divergence in delayed networks, the manuscript strictly bounds the consensus step size (ϵ) by the inverse of the maximum local neighbor degree: $\epsilon \leq \frac{1}{\max(d_i, 1)}$. This formulation is fundamentally incorrect for discrete-time consensus systems experiencing communication delays.

Foundational literature on discrete-time consensus with delays, specifically the seminal work by Olfati-Saber and Murray, explicitly proves that the stability limit for the step size (or coupling gain) in discrete-time systems is strictly bounded by the global maximum degree of the entire network (Δ), not the local degree (d_i). To maintain stability, the Perron matrix defining the state update must remain primitive, and all its eigenvalues must remain within the unit circle on the complex plane. If an agent in a sparse region of the network selects a large step size based only on a small local degree d_i , it creates a localized instability vulnerability. When highly delayed packets from distant parts of the graph eventually arrive, this aggressively large local step size will cause the eigenvalues of the update matrix to exit the unit circle, inducing mean-square state divergence and oscillatory failure across the entire swarm.

Furthermore, robust stability in the presence of communication delays requires the step size to be explicitly constrained by the delay margin itself. For instance, continuous-time equivalent limits require delays to satisfy $\tau < \pi / 2\lambda_N$, a boundary directly related to the maximum network eigenvalues. The manuscript completely lacks a Lyapunov-Krasovskii stability proof, a linear matrix inequality (LMI) condition, or a frequency-domain Nyquist analysis to justify Equation 7 under the severe 90% packet drop and delay constraints claimed in the experiments. The reliance on local degree for bounding delayed discrete-time consensus demonstrates a critical gap in the authors' understanding of distributed control theory.

6. Experimental Validation and Evaluation of Claims

The experimental validation is statistically sound in its methodology, employing 50 independent Monte Carlo runs with randomized initial pseudorandom number generator (PRNG) seeds. Reporting mean metrics alongside 95% confidence intervals demonstrates a clear understanding of empirical stochastic evaluation, separating this work from less rigorous implementations that rely on single-run anecdotal evidence. The ablation studies effectively isolate the performance of the proposed architecture against unconstrained Random Geometric Graph (RGG) baselines and Centralized Oracle bounds.

However, a detailed analysis of the experimental parameters and visual data reveals several discrepancies and unsupported claims.

6.1 Discrepancies in Temporal Scaling

The experimental setup states that the framework was evaluated using 100 autonomous agents running for approximately 285,000 discrete events (ticks). In real-world cyber-physical systems, drone flight controllers and network MAC layers operate at high frequencies, typically between 100 Hz and 1000 Hz. If a single discrete simulation tick represents 10 milliseconds, 285,000 events equate to merely 47.5 minutes of total combined flight time across all 100 agents, which averages to a highly unrealistic 28.5 seconds of real-time operation per drone. Yet, the x-axes on the experimental plots (Figures 3, 5, and 6) show simulation time scaling up to 500 seconds and 300 seconds. This discrepancy requires immediate clarification. Is the DES event queue advancing time non-linearly based on asynchronous event triggering, or is the temporal scale arbitrarily assigned? Without a clear mapping of discrete events to physical time, the kinematic stability and energy decay metrics lose their physical relevance.

6.2 The Missing Energy Cascade

Figure 5 is presented to demonstrate the framework's thermodynamic resilience and prevention of energy cascades, showing a highly stable, perfectly linear energy decay trajectory (0.05 units/tick) even as node attrition reaches 50%. This perfectly linear decay actually undermines the paper's theoretical premise. As discussed in Section 5.3, if the framework accurately models the physics of an RGG, the death of heavily burdened agents should force surviving agents to boost transmission ranges and spatial coverage, which burns exponentially more energy. A robust framework would certainly mitigate this cascade, but the underlying physical stress should still manifest as a slight non-linear curve in the aggregate energy drain as attrition increases. The fact that the energy decay remains flawlessly linear from 100 agents down to 50 agents strongly suggests that the simulator does not actually compute the exponential power requirements of expanded topological bridging. Thus, the claim that the system "effectively prevented inverse energy cascades" is too strong; it appears the simulator simply does not model the cascade mechanics in the physical execution layer.

6.3 Unsupported Claims of Mathematical Bounding

As analyzed extensively, the claim that the Θ_{safe} projector creates a "mathematically bounded stability manifold" is completely unsupported. The empirical evidence in Figure 6 shows that box clamping successfully truncates parameter variance (bounding $\Delta \mu <$

0.04), preventing the massive oscillations seen in the unconstrained heuristic. While empirically successful as a software safeguard, empirical truncation does not equate to a mathematical stability proof. Claiming theoretical bounds based solely on simulation truncation is a severe overstep that will be heavily penalized by academic reviewers.

7. Strongest and Weakest Aspects of the Paper

7.1 Strongest Aspects

1. **The Discrete-Event Simulation Framework:** The engineering effort required to build a deterministically reproducible, discrete-event RGG kernel from scratch is highly commendable. Decoupling the physical simulation from the agent logic to enforce strict epistemic boundaries is a paradigm that many professional simulation environments fail to implement correctly. The architecture is robust, scalable, and thoughtfully designed.
2. **Rigorous Ablation Methodology:** The quantitative ablation study in Table II and Figure 4 clearly isolates the performance metrics. Comparing the proposed architecture against both an unconstrained baseline (to show improvement) and a centralized oracle (to show the theoretical upper bound) is the gold standard for experimental design in systems engineering.
3. **Writing Quality, Narrative Flow, and Data Visualization:** The manuscript is remarkably well-written. It utilizes precise academic vocabulary, clear signposting, and fluid transitions between complex topics. The problem is sharply motivated, and the figures (e.g., Figures 3, 4, and 6) are of extremely high publication quality, effectively communicating complex multi-dimensional data.

7.2 Weakest Aspects

1. **Absence of Distributed Estimator Dynamics:** The failure to provide the distributed mathematical equations for estimating the Fiedler value (λ_2) while simultaneously claiming its use in a strictly localized fog-of-war environment is a fatal contradiction.
2. **Ignorance of Delay-Dependent Stability Margins:** The use of localized degree to bound discrete-time consensus step sizes in a highly delayed network ignores two decades of foundational control theory literature. The lack of Lyapunov proofs or Nyquist frequency-domain analysis renders the control algorithms theoretically invalid.
3. **Overstated Theoretical Claims:** The persistent characterization of simple software engineering techniques (box clamping, linear subtraction) as profound mathematical stability manifolds or thermodynamic cascade models demonstrates a lack of maturity in theoretical formalization.

8. Academic Maturity and Baseline Comparison

The manuscript reads like the work of an exceptionally ambitious, early-stage Master's student or a top-tier undergraduate capstone project.

It does not read like an early PhD research paper submitted to a venue like IEEE Transactions on Automatic Control. A PhD-level paper in this domain would focus heavily on the rigorous mathematical proofs supporting the distributed algorithms. It would provide theorems, lemmas, and Lyapunov stability criteria proving that the proposed discrete-time consensus and

adaptation algorithms converge under specified delay margins and packet drop probabilities. The total absence of strict mathematical proofs and the reliance on heuristic box-clamping reveals a lack of formal graduate-level training in nonlinear control theory and distributed optimization.

However, when evaluated objectively against typical final-year undergraduate projects at a tier-3 engineering college in India, this work is **significantly above average, bordering on exceptional**. The vast majority of tier-3 computer science capstone projects heavily rely on implementing off-the-shelf machine learning APIs (e.g., applying pre-trained YOLO models to generic image datasets) or building basic web and mobile applications using standardized frameworks. They rarely, if ever, venture into the complexities of multi-agent systems, stochastic graph theory, spectral connectivity, or deterministic discrete-event simulation.

The fact that this student has independently navigated complex, graduate-level domain literature—spanning spectral graph theory, percolation thresholds, and consensus dynamics—and synthesized it into a bespoke, working software architecture without experienced faculty guidance is a massive achievement. While the mathematics are flawed by professional IEEE standards, the mere attempt to implement localized SSI auctions and parameter projection within an epistemically constrained C++/Python environment places the author in the top one percent of their immediate peer group. The conceptual synthesis, even if theoretically imperfect, demonstrates profound systems-engineering capability.

9. Impression for MS Admissions (CS, AI, Robotics, CPS)

If an admissions committee for a top-tier Master's program in Computer Science, Artificial Intelligence, Robotics, Distributed Systems, or Cyber-Physical Systems were evaluating this paper in a portfolio, it would leave a highly favorable and lasting impression.

Admissions professors are fully aware that undergraduate students rarely produce theoretically flawless research. Instead, they look for strong evidence of independent research capability, advanced systems building skills, intellectual curiosity, and an understanding of the scientific method (such as utilizing baselines, ablation studies, and statistical confidence intervals). This manuscript provides all of these indicators in abundance.

An admissions professor reading this would immediately recognize the mathematical gaps regarding distributed spectral estimation and delay-dependent stability bounds. However, they would appropriately attribute these flaws to the applicant's lack of formal graduate coursework in nonlinear control and advanced dynamic systems. The paper clearly signals that the applicant has exceptional software engineering foundations, a firm grasp of the macroscopic challenges in decentralized AI, and the ambition to tackle mathematically dense topics independently. They would view this student as highly teachable, technically proficient, and possessing immense potential for success in a rigorous, research-intensive graduate laboratory. For a program heavily focused on applied systems building, this paper is a tremendous asset.

10. Venue Suitability and Publication Readiness

If this manuscript were submitted today without any modifications, its estimated chances for various academic venues are detailed below:

Venue Category	Estimated Chance	Detailed Justification
Student Research Conference	Accept (95%)	The monumental engineering effort, exceptionally clear writing, excellent data visualization, and novel simulation architecture make this an outstanding student paper. It easily exceeds the technical and presentation thresholds for venues aimed specifically at undergraduates and early-stage graduate students.
Regional IEEE Conference (e.g., IEEE INDICON, Region 10)	Accept (75%)	Regional IEEE conferences generally prioritize clear system implementations, practical engineering applications, and solid empirical results over rigorous mathematical proofs. The comprehensive ablation studies, statistical rigor, and professional presentation will satisfy the majority of reviewers, who are likely to overlook or forgive the missing mathematical stability proofs given the strength of the software implementation.
ICRA / IROS (Premier Robotics Conferences)	Reject (10%)	These premier venues demand highly mature research consisting of either (a) rigorous theoretical contributions with extensive mathematical proofs of stability and convergence, or (b) extensive physical hardware validation (Hardware-in-the-Loop or deployment on physical UAVs). This paper relies purely on a synthetic 2D grid simulation and heuristic mathematics, falling significantly short of the demanding standards of top-tier international robotics conferences.
IEEE Robotics and Automation Letters (RA-L)	Reject (0%)	RA-L is a premier journal that requires highly original,

Venue Category	Estimated Chance	Detailed Justification
		mathematically proven, and heavily validated research. The use of simple box clamping masquerading as a "stability manifold," the fundamentally incorrect local degree bounds for delayed discrete-time consensus, and the total absence of the distributed λ_2 estimator equations would trigger an immediate desk rejection by the editorial board.

11. Quantitative Evaluation Rubric (0 – 10)

Category	Score	Rationale
Novelty	6.0	The simulation architecture enforcing the epistemic boundary is conceptually fresh and practically valuable, but the underlying control algorithms (box clamping, SSI auctions, regime switching) are highly derivative and lack fundamental innovation.
Technical Depth	7.5	Outstanding software architecture, robust event-queue management, and deep integration of localized constraints. However, it loses points for lacking depth in the algorithmic handling of asymmetric topologies and non-linear physical dynamics.
Mathematical Rigor	3.0	The lowest scoring category. Heuristic parameter bounding is falsely equated to a mathematical stability manifold. Completely missing distributed estimation equations for the Fiedler value. Incorrect bounds for discrete-time consensus in delayed networks.
Experimental Validation	7.5	Highly commendable, statistically rigorous use of Monte Carlo runs and

Category	Score	Rationale
		confidence intervals. Points deducted because temporal scales (ticks vs. seconds) are misaligned, and the perfectly linear energy decay directly contradicts the paper's "energy cascade" hypothesis.
Writing Quality	9.0	Exceptionally articulate, well-structured, and persuasive. The author uses precise domain vocabulary effectively and maintains a highly professional, academic tone throughout the manuscript.
Organization	8.5	Logical, seamless flow from problem motivation to system architecture, mathematical modeling, and finally empirical ablation studies.
Figures and Tables	8.0	Clean, highly professional architecture diagrams and data plots. Visuals effectively support and enhance the textual narrative.
Reproducibility	9.5	The commitment to deterministic PRNG seeds, independent random number streams, and an open-source GitHub release constitutes the gold standard for simulation-based research.
Overall Publication Readiness	6.0	Ready for regional conferences or student symposiums immediately. Requires massive mathematical revisions and theoretical formalization before submission to any high-impact journal or premier conference.

12. Formal Program Committee Review (Reviewer #2 Mock Review)

To: The Authors From: Program Committee Reviewer #2 Venue: IEEE International Conference on Robotics and Automation (ICRA) Recommendation: Weak Reject

Summary

The manuscript proposes a discrete-event simulation (DES) framework designed to evaluate decentralized multi-agent drone swarms operating under strict "Fog-of-War" constraints. These constraints include stochastic packet dropping, asynchronous communication delays, and thermodynamic energy depletion. The authors introduce a Hybrid Supervisor that utilizes a Θ_{safe} box-clamping projector to bound heuristic parameter adaptation, allegedly preventing kinematic instability. The framework is validated via a simulation of 100 agents over 285,000 discrete events, with results demonstrating preserved spectral connectivity and linear energy decay when compared to unconstrained baseline models.

Major Strengths

1. **Rigorous Simulation Architecture:** The strict decoupling of the deterministic discrete-event kernel from the decentralized agent logic is highly commendable. By architecturally preventing agents from accessing the global adjacency matrix, the framework enforces a true epistemic boundary. This is a significant improvement over contemporary continuous-time consensus simulations that inadvertently leak global state data, providing an excellent testbed for realistic algorithm evaluation.
2. **Empirical Methodology and Reproducibility:** The use of 50 independent Monte Carlo runs with fixed PRNG seeds to generate 95% confidence intervals across ablation studies demonstrates a rigorous approach to stochastic evaluation. Furthermore, the commitment to open-sourcing the deterministic simulation kernel sets a strong standard for reproducibility.
3. **Clarity of Exposition:** The manuscript is exceptionally well-written and organized. The research problem is sharply motivated, the literature gap is accurately identified, and the complex data pipeline (Figure 1) is clearly illustrated.

Major Weaknesses

1. **Missing Distributed Estimator Dynamics for Algebraic Connectivity:** The manuscript makes a foundational claim that agents autonomously trigger regime shifts based on local proxies of the Fiedler value (λ_2). However, Equation 2 merely provides the centralized Rayleigh quotient definition of algebraic connectivity. The Fiedler value is a global property of the graph Laplacian. Generating a local proxy in a highly dynamic, delayed, and stochastically dropping network requires complex distributed estimation algorithms (e.g., decentralized power iteration, PI consensus tracking, or distributed observers). The complete absence of this estimator implies that the DES kernel might be calculating λ_2 globally and feeding it to the agents, which directly violates the paper's core "Fog-of-War" constraint.
2. **Flawed Consensus Stability Bounds for Delayed Networks:** Equation 7 defines the asynchronous step size as $\epsilon \leq 1/\max(d_i, 1)$, relying solely on local neighbor degree. However, established literature on discrete-time consensus dictates that in the presence of communication delays, the step size must be bounded by the global maximum degree of the network (Δ), and constrained by the maximum time delay margin, to ensure the Perron matrix remains stable and eigenvalues remain within the unit circle. A locally computed step size in a highly delayed RGG will inevitably lead to

mean-square state divergence under specific boundary conditions. The manuscript provides no Nyquist criteria, LMI conditions, or Lyapunov analysis to prove stability under these delayed conditions.

3. **Mischaracterization of Box Clamping as a Stability Manifold:** Section IV characterizes static box clamping (Equation 4) as a "stability-constrained" projection onto a "mathematically bounded stability manifold." This is a severe theoretical overstatement. Clamping a heuristic parameter to arbitrary minimum and maximum values does not ensure closed-loop dynamical stability; it merely prevents integer/float overflow or extreme kinematic inputs. True safe-set projection in modern control theory requires evaluating the control input against the gradient of a Control Barrier Function (CBF) or utilizing invariant set theory.

Required Revisions for Resubmission

To meet the threshold for acceptance at a venue of this caliber, the authors must address the following fundamental gaps:

1. **Formalize the λ_2 Estimator:** Explicitly provide the distributed update difference equations used by individual agents to estimate λ_2 . If a standard estimator from literature is utilized, cite it directly and explain how its convergence properties are maintained under the severe 90% Bernoulli packet drop rate modeled in Section V.
2. **Revise Consensus Stability Limits:** Modify Equation 7 to account for time-delay stability margins. Provide a brief theoretical proof, or directly cite appropriate literature (e.g., Olfati-Saber & Murray), demonstrating that the chosen step size guarantees discrete-time mean-square stability despite asynchronous, non-uniform packet delays.
3. **Reframe Θ_{safe} :** Downgrade the claims regarding the mathematical stability manifold. Accurately describe the box clamping and EMA pipeline as a heuristic kinematic safeguard rather than a theoretically bounded control manifold, or replace Equation 4 with a true gradient-based safety filter.

Minor Comments

- **Asymmetric Voronoi Overlap:** In an RGG with random packet drops, the adjacency matrix is inherently asymmetric ($A_{ij} \neq A_{ji}$). Consequently, local Voronoi cells (Equation 6) computed by neighboring agents under mismatched information will conflict and overlap. How does the SSI auction or collision avoidance system resolve these overlapping territorial claims?
- **Simulation Timescale Discrepancy:** Section V.A notes testing over 285,000 discrete events, yet the time axes in Figures 3, 5, and 6 span from 200 to 500 seconds. Please explicitly define the temporal resolution of a simulation "tick" and clarify how event density maps to simulated physical seconds.
- **Energy Model Contradiction:** Equation 3 relies on a purely linear energy decay function. In reality, transmission power scales non-linearly with distance, and battery discharge is non-linear. The authors should comment on how a linear mathematical model can accurately capture the exponential "inverse energy cascade" claimed in the introduction, particularly since Figure 5 displays a perfectly linear decay curve even at 50% node attrition.

Final Recommendation

Weak Reject. While the simulation engineering, ablation methodology, and written presentation are of very high quality, the theoretical formulations underlying the control algorithms are fundamentally incomplete. The omission of the distributed Fiedler value estimator, the incorrect delay-consensus bounds, and the overstatement of the box-clamping mechanism render the core control-theoretic claims mathematically unsupported. I strongly encourage the authors to integrate formal distributed estimation dynamics and rigorous stability proofs, and resubmit this promising framework to a future conference.

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